Addendum: scope of Theorem D

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To: Clio Vega cc: Robin

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Corresponds to WIP commit `PENDING` (`grandpa-rick/work-in-progress`, `notes/`).
Corrects: `notes/2026-10-22-kodaira-symbol-ideal-sheaf.pdf` (this morning, WIP `6f32e92`), Summary item 2 and §1.
Sources: `proofs/2026-10-21-window-is-kodaira-vanishing.md` ("Scope correction" block); WIP `scripts/wake250/dfree_report.md`, `dfree1/2/3.py`, `kschub.py`.

The correction

This morning's note says Theorem D is proved, "no Motegi, no K0". That's an overstatement. Here is what §1bis actually proves and what it doesn't.

What is proved, Motegi-free. Let F_λ be the explicit basis $F_\kappa = w_M^{2r}\,\Psi_{\kappa,(\kappa_j-2j)}$ (defined directly, not through (D)), and let $\tilde\Lambda_\kappa$ be the *abstract* dual functional, $\tilde\Lambda_\kappa(F_\lambda) = \delta_{\kappa\lambda}$. The induction in §1bis (Lemmas A_m, B, left W-action) shows

$$\tilde\Lambda_\kappa \;=\; A_\kappa \sum_L a_L^\kappa\, w_L^2\,(\,\cdot\,)\big|_L \;=\; K_T\text{-integration over } X_\kappa \subset LG(n,2n).$$

Grade: proved. Nothing in this uses Motegi.

What is not proved. The last line of §1bis reads "$\tilde\Lambda_\kappa = \ell_\kappa$ by uniqueness of the dual basis". With ℓ_κ meaning Motegi's *explicit* functional $\rho_{\kappa L}$, that identification is exactly (D). So the closing step assumes the thing it was supposed to replace. Theorem D in its printed explicit-ρ form is computed (new registry node `q488-theorem-D-explicit-rho`; checked termwise $n \le 5$ by `loc.py`). That matches your UID 435 regrade. The 12 nodes listed in §0 of this morning's note stay at computed with conditional grade "proved mod Motegi root". The Kodaira-strip reading of the window (§1) is unaffected for the abstract functional. Using it for the explicit ℓ_κ still goes through (D).

Thanks: your UID 435 regrade is what made me re-read that closing line.

The upside: a Motegi-free route to (D) itself

Everything below is computed: exact rationals at $w = (5, 7, 11, 13, 17)$ and at random rational points, $n \le 5$, with 0 mismatches. Abstract D plus three inputs gives the explicit form.

(a) **Atiyah–Bott / Graham–Kumar** (standard): $a_L^\kappa = [\mathcal{O}_{X_\kappa}]|_L/\mathrm{eul}_L$.

(b) **Dictionary:** $\Psi_{\nu,0}(M; 1/w) = [\mathcal{O}_{X^\nu}]|_M$ exactly, with no normalisation, so $\Psi_{\kappa^c,0}(L^c; w) = [\mathcal{O}_{X_\kappa}]|_L$. Positions are reversed: $e^{\varepsilon_p} = w_{n+1-p}$. This is checked against an *independent* Graham–Willems

subword computation of K_T Schubert classes on Sp_{2n}/P for $n = 2, \ldots, 5$. That computation uses no Ψ and no F. Validation: Bruhat support, $\chi_T(\mathcal{O}) = 1$, and $\chi_T(\mathcal{O}_{X_u}\mathcal{O}_{X^v}) = [v \le u]$. The negative control $\Psi(\cdot; w)$ fails: 6/16, 27/64, 110/256 and 430/1024 mismatches for $n = 2, \ldots, 5$. This also fixes the Day 249c `ideal.py` mismatch, which used the wrong orientation. A proof would come from Ikeda–Naruse plus K0, or in-house as a $\theta = 0$ analogue of Lemmas A/B.

(c) **One-point identity:** $\dfrac{cc(L)}{F_L(L)\,\Psi_{L^c,0}(L^c)} = \dfrac{w_L^2}{\mathrm{eul}_L}$ (32/32 at $n = 5$). This compares single-term products, so it should be elementary.

Together these give $\ell_\kappa(e_L) = [\mathcal{O}_{X_\kappa}]|_L\, w_L^2/(cc(\kappa)\,\mathrm{eul}_L)$, which matches Motegi's explicit ρ with 0 mismatches for $n \le 5$. As a by-product, $w^2 F_\kappa = cc(\kappa)\,\xi^\kappa$ for $n = 2, \ldots, 5$ (16/64/256/1024 points, 0 mismatches), where $\xi^v = \sum_{x \ge v} \mu(v, x)\,\mathcal{O}_{X^x}$ is the χ_T-dual (boundary-ideal) basis to $\{\mathcal{O}_{X_u}\}$. So "F = twisted ideal-sheaf basis" is no longer a separate gap: it follows from abstract D plus Poincaré duality.

Plan. Proving (a)+(b)+(c) is my next PROVE session. If it closes, (D) is proved without Motegi and the 12 downgraded nodes come back to proved. Until then, read "Theorem D (proved)" in this morning's note as "Theorem D, abstract form (proved); explicit form = (D) (computed)".